Missions planned to enable stages of reproduction beyond Earth

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CEO SpaceBorn United

FISO Telecon 08-28-2024

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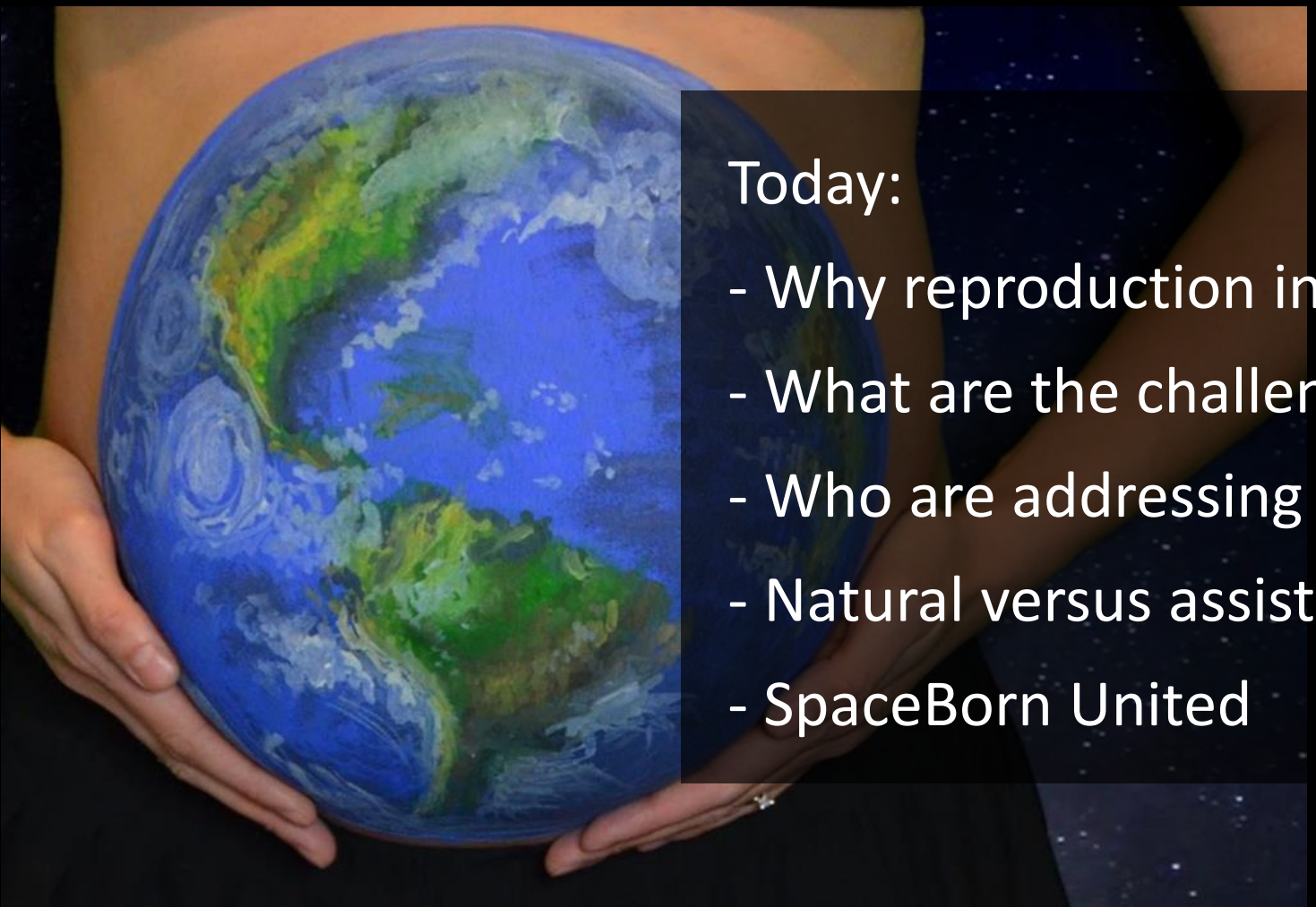
Missions planned to enable stages of reproduction beyond Earth



Today:

- Why reproduction in space?
- What are the challenges?
- Who are addressing them?
- Natural versus assisted reproduction
- SpaceBorn United

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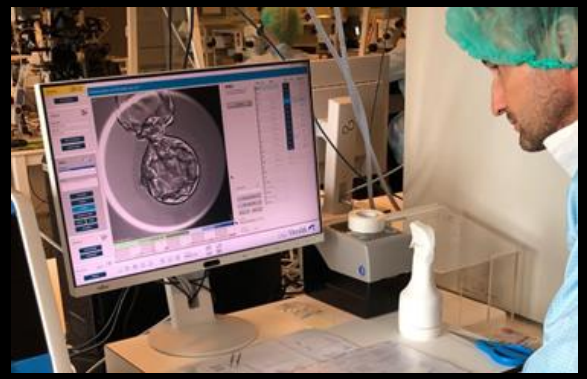
Why reproduction in space?

- Independent settlements beyond Earth
- Improve reproductive biology on Earth
- Geo-political prestige upcoming space nations
- Address uncontrolled conception in Space Tourism



Why reproduction in space?

Independent settlements beyond Earth:

- Exploration is in our DNA
- Protecting future generations
- Enriching life in space and on Earth

Why reproduction in space? II

- Shorter IVF treatments
- Reduced medication costs
- Improved embryo selection
- Improved IVF culture medium
- Reduced patient burden

IVF in space improves IVF success rates on Earth

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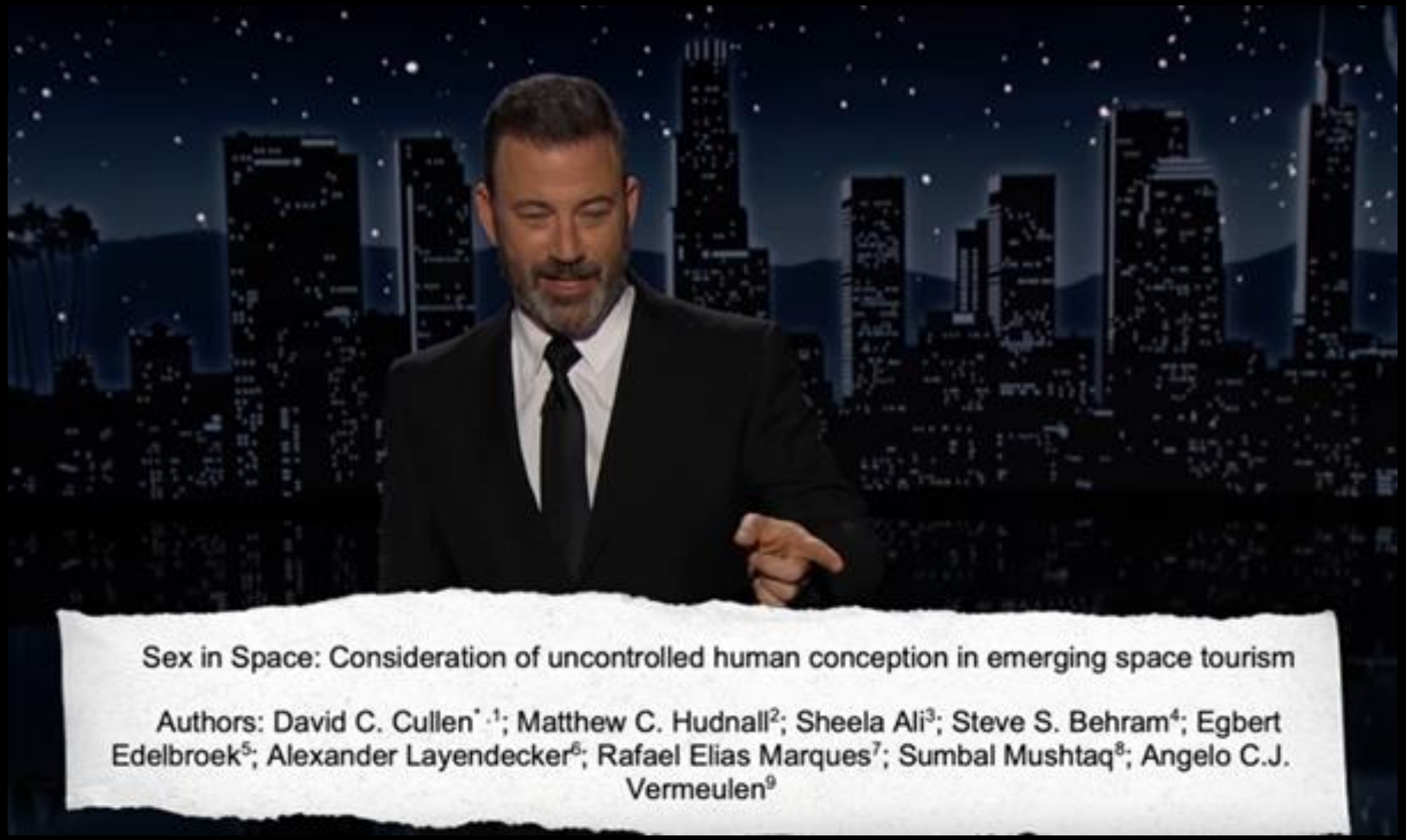
Why reproduction in space? III

Geo-political prestige upcoming space nations

Space Tourism: a magnet for Sex in Space



Reproduction in space

What are the challenges?

Who are addressing them?

Defining a roadmap

Natural or artificial (assisted) reproduction in space?

Re-engineering existing A.R.T.

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Shaping the Future of Reproductive Biology on Earth and in Space

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What do we do?

1. Research conditions for safe stages of human reproduction in space

2. Translate research outcomes to missions designs (+ exec. partners)

3. Translate research outcomes into required biomedical equipment

ARTIS™

Assisted Reproductive Technology In Space

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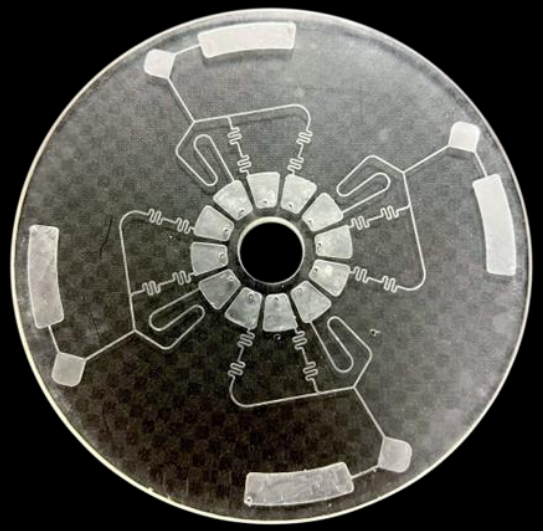
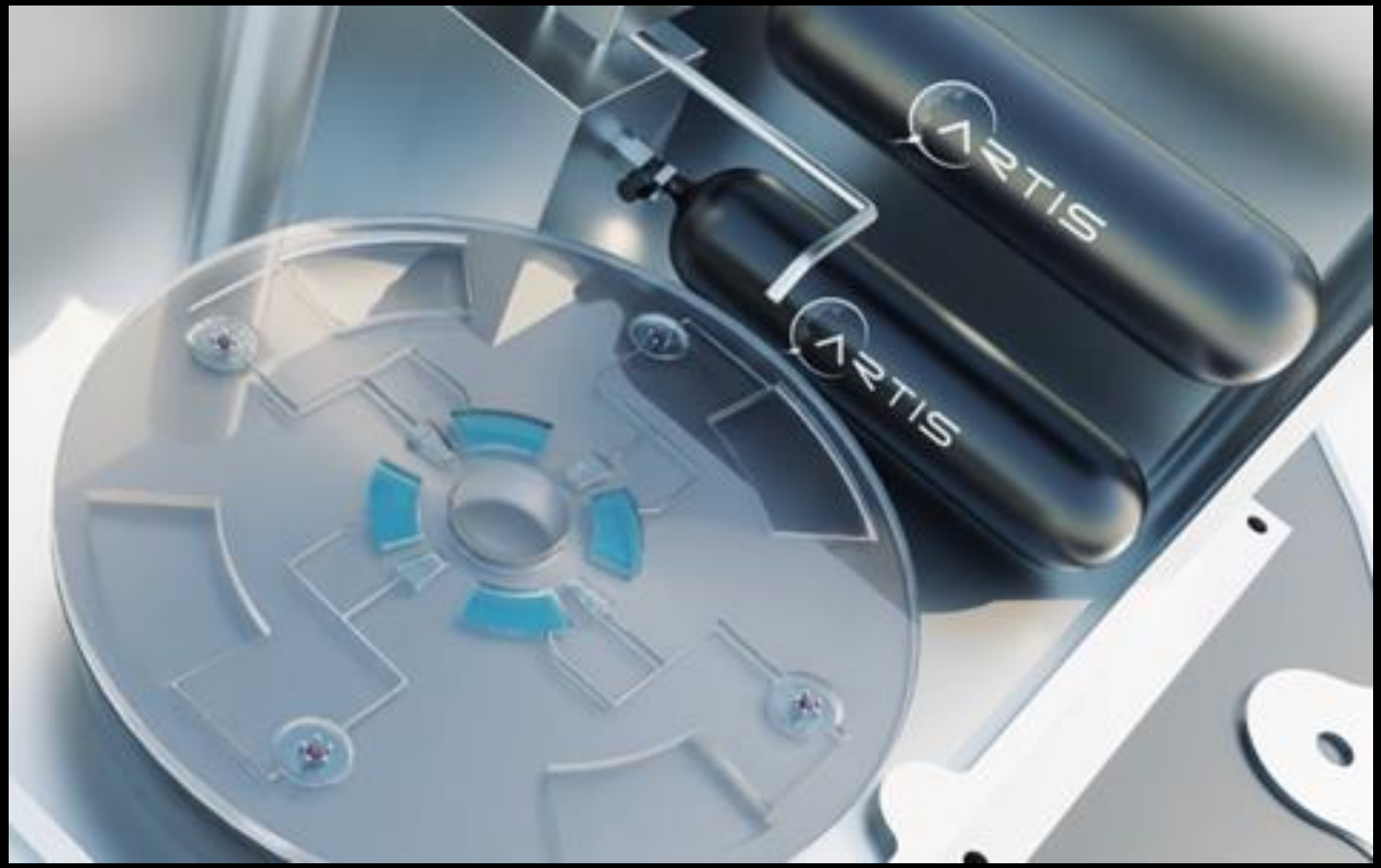
Assisted Reproductive Technology In Space





Research & Industry Partners

Frontier Space Technologies

PROGENESIS
We help you start a healthy family

PRECIOUS PAYLOAD

OFFWORLD BIOTECH

Vitrolife

BiologIC Technologies

Rapid Fluidics
Shaping the Future of Microfluidics

GHENT UNIVERSITY

SPACETECH ANALYTICS

PARAGON
SPACE DEVELOPMENT CORPORATION

INDEPENDENCE X
AEROSPACE

spaceapplications SERVICES

TU/e EINDHOVEN UNIVERSITY OF TECHNOLOGY

HARVARD UNIVERSITY

Cranfield UNIVERSITY

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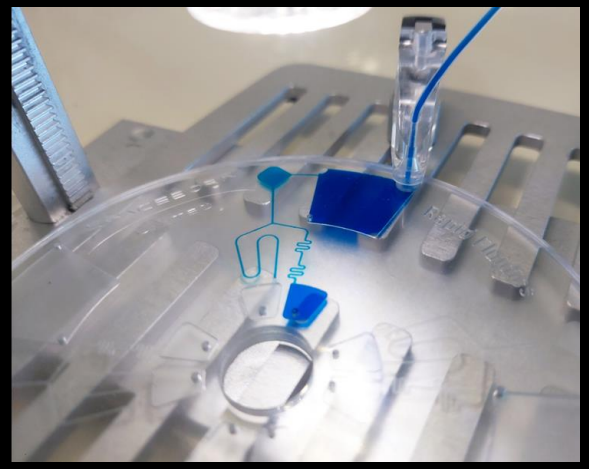

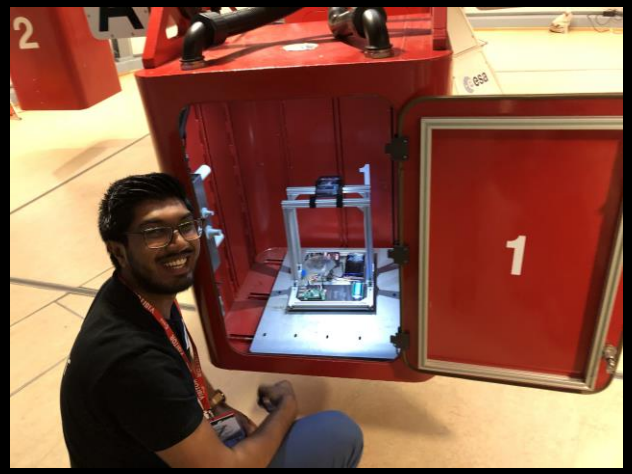

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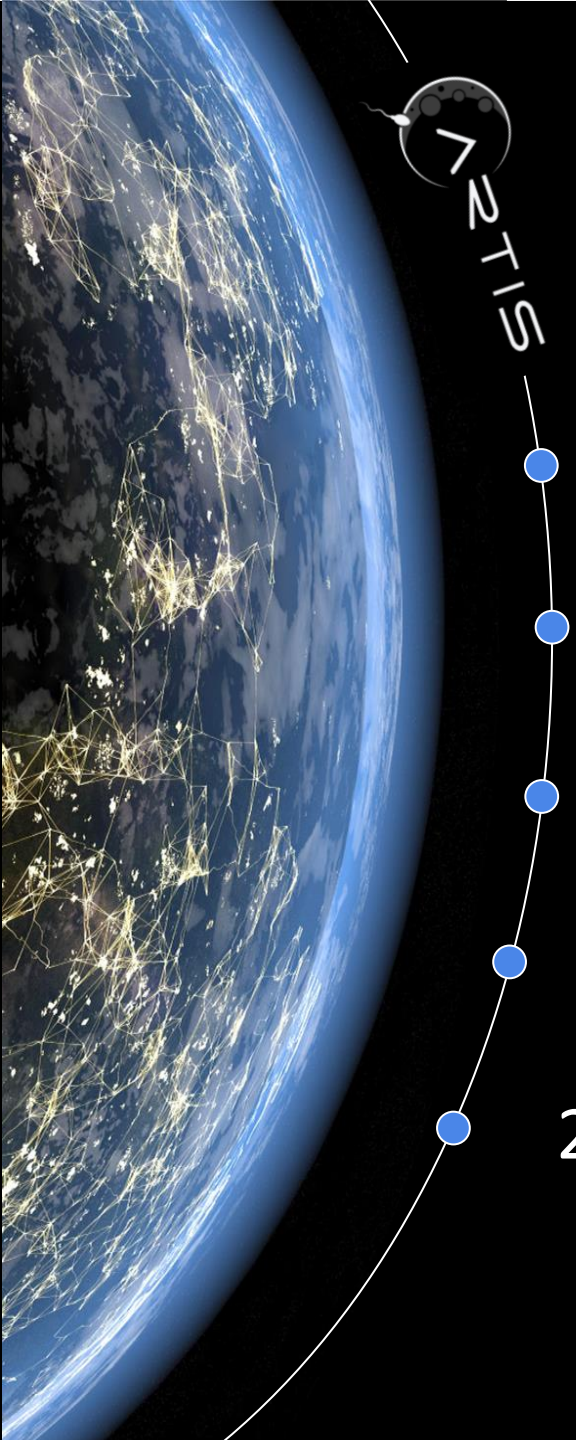
Roadmap

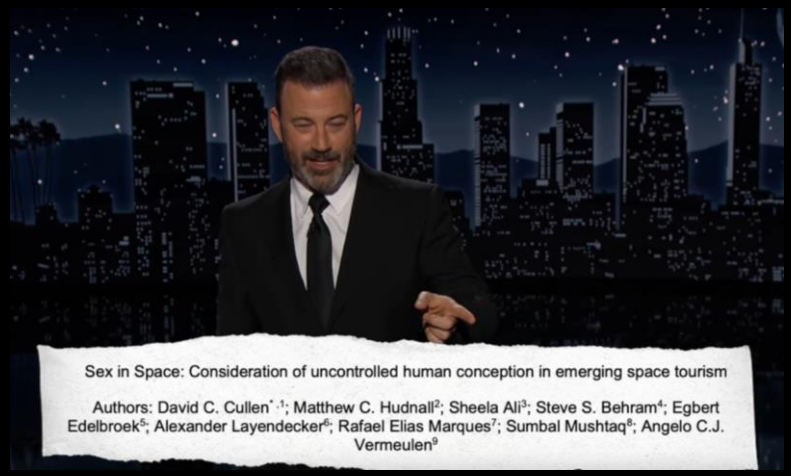


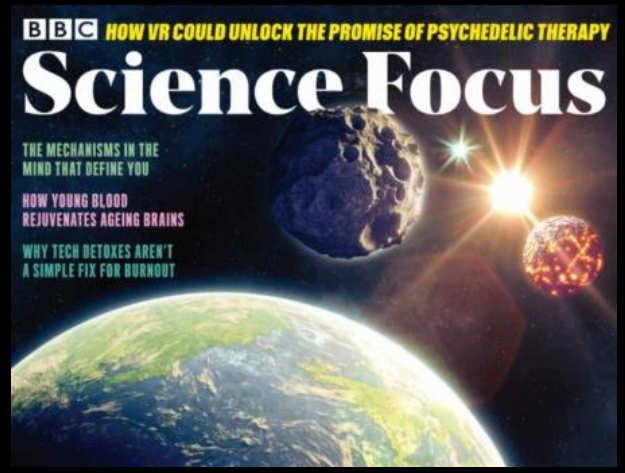
- 2024 Ground based testing campaign
- 2025 ARTIS prototype validation in space (2x)
- 2026 MVP: full 5 day IVF cycle in RPM and in space
- 2027 IVF cycle + sample recovery to Earth
- 2030 1st Human baby conceived in space

High Altitude (24 km) Drop Test

Aug. 2023

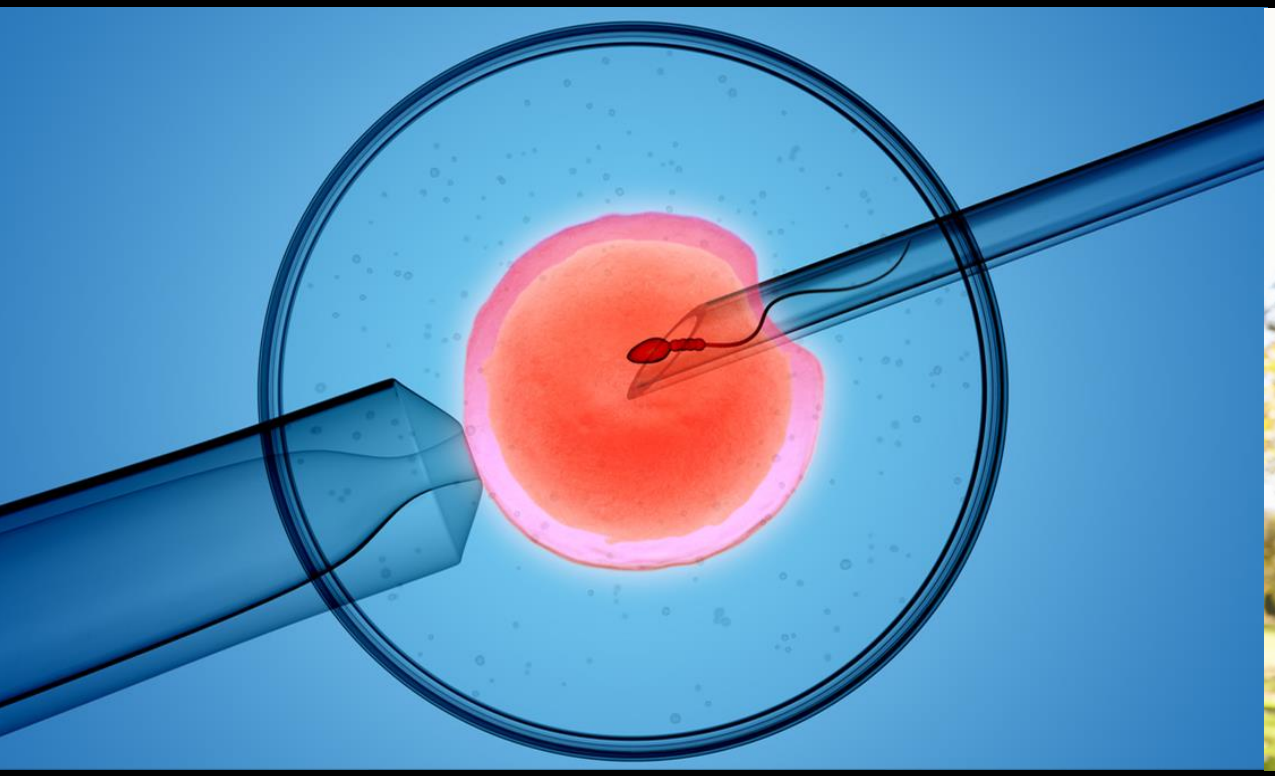
Media & Science publications





Shaping the Future of Reproductive Biology

On Earth: Improving IVF succes

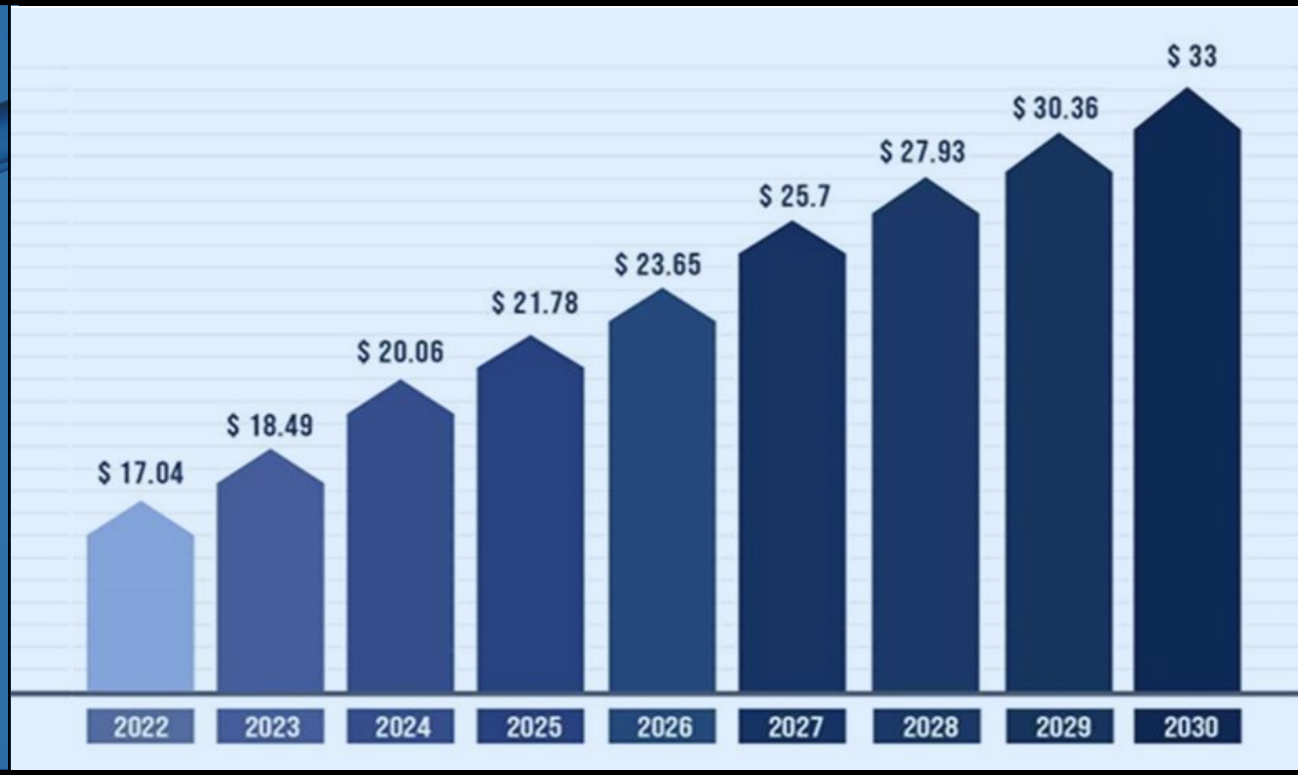


IVF in space improves IVF success rates on Earth

- **Shorter IVF treatments**
- **Reduced medication costs**
- **Improved embryo selection**
- **Improved IVF culture medium**
- **Reduced patient burden**

Strong Fertility / IVF market opportunities

- 1 in 6 couples need fertility care
- Fertility rates keep decreasing
- Women start families later
- Demand outpacing supply
- IVF success rates are stagnant
- 2.5 million IVF cycles per year



IVF (in vitro fertilization) market size 2020 - 2030 ($ BN)

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Our Competitive Advantage

	overture	conceivable	TMRW	AUTOIVF BIOTECH	fertilis	IVF 2.0	SPACEBORN UNITED
Space benefits							✓
Automation	✓	✓	✓	✓		✓	✓
A.I.		✓		✓		✓	✓
Cryogenic Innovation		✓	✓		✓		✓

Thank you!

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Spreading the seeds of life on Earth and beyond

Expanding the human comfort zone